Hao Li

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Education

University of Maryland, College Park — College Park, MD
Ph.D. in Computer Science — 2012 – 2018
- Advisors: Prof. Hanan Samet and Prof. Tom Goldstein
- Dissertation: Towards Fast and Efficient Representation Learning

Institute of Computing Technology, Chinese Academy of Sciences — Beijing, China
M.S. in Computer Science — 2009 – 2012
- Advisor: Prof. Shiguang Shan

Shandong University — Jinan, China
B.Eng. in Software Engineering — 2005 – 2009

Research Interests

Machine Learning, Deep Learning, Computer Vision, Distributed Computing

Work Experience

Amazon Web Services — Seattle, WA
Applied Scientist II with AWS Rekognition Team — Oct 2018 - present
- Research and develop machine learning and computer vision algorithms and models for AWS image and video analytical services, including Content Moderation and Custom Labels.
- Research on efficient hyperparameter optimization for transfer learning [ICLR'20].

University of Maryland Institute for Advanced Computer Studies — College Park, MD
Research Assistant with Prof. Hanan Samet and Prof. Tom Goldstein — May 2013 - Aug 2018
- Visualizing the loss surface of neural networks [NeurIPS'18]
- Optimization methods for training quantized neural networks [NIPS'17].
- Accelerating Convolutional Neural Networks for fast inference [ICLR'17].
- Distributed machine learning with HPC [EuroSys'15, ICML'17].

Machine Learning Team, Nervana Systems — San Diego, CA
Algorithms Engineer Intern with Scott Cyphers and Arjun Bansal — May - Aug 2016
- Design and implement the importer module for Neon's graph-backend (Nervana Graph), enabling executing the computation graph of TensorFlow with nGraph.

Machine Learning Department, NEC Labs America — Princeton, NJ
Research Intern/Visiting Student — May - Aug'15/Jan - May'16
- Research on the architecture design of Convolutional Neural Networks.
- Research on filter pruning for acceleration and compression of CNNs [ICLR'17].

Storage System Department, NEC Labs America — Princeton, NJ
Research Intern with Asim Kadav, Erik Kruus and Cristian Ungureanu — May – Aug 2014
- Built MALT, a distributed data-parallel machine learning system with flexible peer-to-peer communication, utilizing one-side RDMA write over InfiniBand [EuroSys'15].

Institute of Computing Technology, Chinese Academy of Sciences Beijing, China
Research Assistant with Prof. Shiguan Shan Jan 2011 - July 2012

– Multi-person tracking with re-identification in non-overlapping camera networks [ACCV'14].

Internet Graphics Group, Microsoft Research Asia Beijing, China
Research Intern with Ying-Qing Xu May – Nov 2010

– Bing image search result beautification.

Media Computing Group, Microsoft Research Asia Beijing, China
Research Intern with Meng Wang Mar – July 2009

– Research on graph-based reranking for web image search [T-IP'12, SIGIR'11].

– Built a million-scale image/video dataset for web image/video search reranking [ICDMW'09].

Publications

1. *Rethinking the Hyperparameters for Fine-tuning*
Hao Li, Pratik Chaudhari, Hao Yang, Michael Lam, Avinash Ravichandran, Rahul Bhotika, Stefano Soatto, International Conference on Learning Representations (**ICLR**), 2020

2. *Visualizing the Loss Landscape of Neural Nets*
Hao Li, Zheng Xu, Gavin Taylor, Tom Goldstein
Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2018

3. *Training Quantized Nets: A Deeper Understanding*
Hao Li*, Soham De*, Zheng Xu, Christoph Studer, Hanan Samet, Tom Goldstein
Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2017
ICML Workshop on Principled Approaches to Deep Learning, 2017 (**Google Best Student Paper**)

4. *Pruning Filters for Efficient ConvNets*
Hao Li, Asim Kadav, Igor Durdanovic, Hanan Samet, Hans Peter Graf
International Conference on Learning Representations (**ICLR**), 2017
NIPS Workshop on Efficient Methods for Deep Neural Networks (**EMDNN**), 2016

5. *Adaptive Consensus ADMM for Distributed Optimization*
Zheng Xu, Gavin Taylor, **Hao Li**, Mario Figueiredo, Xiaoming Yuan, and Tom Goldstein
International Conference on Machine Learning (**ICML**), 2017

6. *MALT: Distributed Data-Parallelism for Existing ML Applications*
Hao Li, Asim Kadav, Erik Kruus, Cristian Ungureanu
ACM European Conference on Computer Systems (**EuroSys**), 2015
NIPS Workshop on Machine Learning Systems (**LearningSys**), 2015

7. *Streaming News Image Summarization*
Hao Li, Shangfu Peng, Hanan Samet
International Conference on Pattern Recognition (**ICPR**), 2016 (Oral)

8. *Simplification and Refinement for Speedy Spatio-temporal Hot Spot Detection Using Spark (GIS Cup).*
Shangfu Peng, Hong Wei, **Hao Li**, and Hanan Samet
ACM Int. Conf. on Advances in Geographic Information Systems (**SIGSPATIAL**), 2016 (Short)

9. *View Adaptive Metric Learning for Multi-view Person Re-identification*
Canxiang Yang, Shiguang Shan, Dan Wang, **Hao Li**, Xilin Chen
Asian Conference on Computer Vision (**ACCV**), 2014

10. *Multimodal Graph-Based Reranking for Web Image Search*
Meng Wang, **Hao Li**, Dacheng Tao, Ke Lu, Xindong Wu
IEEE Transactions on Image Processing (**T-IP**), 2012

11. *Optimizing Multimodal Reranking for Web Image Search*
Hao Li, Meng Wang, Zhisheng Li, Zheng-Jun Zha, Jialie Shen
ACM SIGIR Conf. on Research and Development in Information Retrieval (**SIGIR**), 2011 (Short)

12. *MSRA-MM 2.0: A Large-Scale Web Multimedia Dataset*
Hao Li, Meng Wang, Xian-Sheng Hua
IEEE ICDM Workshop on Internet Multimedia Mining, 2009

Awards

Google Best Student Paper Award, ICML PADL Workshop	2017
Conference Travel Awards: NIPS'17, ICML'17, ICLR'17, EuroSys'15, SIGIR'11	
Jacob K. Goldhaber Travel Grant, University of Maryland	2016
John D. Gannon Student Travel Grant, University of Maryland	2013
Dean's Fellowship, Department of Computer Science, University of Maryland	2012, 2013
Stars of Tomorrow Excellent Intern Award, Microsoft Research Asia	2010
Outstanding College Graduate, Shandong Province and Shandong University	2009
Meritorious Winner in Mathematical Contest in Modeling, USA	2008
National Second Prize in China Undergraduate Mathematical Contest in Modeling	2007

Academic Services

Editorial board reviewer for Journal of Machine Learning Research (JMLR)
Journal reviewer for International Journal of Computer Vision (IJCV)
Journal reviewer for IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)
Journal reviewer for IEEE Transactions on Neural Network and Learning Systems (T-NNLS)
Journal reviewer for IEEE Transactions on Multimedia (T-MM)
Journal reviewer for IEEE Transactions on Industrial Informatics
Journal reviewer for Neurocomputing
Journal reviewer for Multimedia Systems
Reviewer for Neural Information Processing Systems (NeurIPS)
Reviewer for International Conference on Machine Learning (ICML)
Reviewer for International Conference on Learning Representation (ICLR)
Reviewer for IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
Reviewer for International Conference on Computer Vision (ICCV)
Reviewer for European Conference on Computer Vision (ECCV)
Reviewer for ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
Reviewer for AAAI Conference on Artificial Intelligence (AAAI)

Talks

Towards Fast and Efficient Representation Learning

- Facebook Research, Menlo Park, CA, May, 2018
- IBM Research, Cambridge, MA, May, 2018
- Amazon AWS AI, Seattle, WA, April, 2018
- Apple AI Research, Cupertino, CA, April, 2018

- Vicarious, Union City, CA, April, 2018
- Comcast Labs, Washington D.C., Mar, 2018
- Microsoft Research Asia, Bejing, Jan. 2018
- NVIDIA Research, Santa Clara, CA, Nov. 2017

Towards a Deeper Understanding of Training Quantized Neural Networks

- ICML'17 Workshop on Principled Approaches on Deep Learning, Sydney, Aug. 2017

Programming Skills

Languages: Python, C/C++, Lua, Java, C#, Matlab, Go
Frameworks: PyTorch, MXNet, TensorFlow, Torch7, Neon, MPI, GASPI, ZMQ, Hadoop

Patents

Parallelized machine learning with distributed lockless training. US Patent 9,984,337
Passive Pruning of Filters in a Convolutional Neural Network. US Patent App. 15/590,620
Security System Using a Conv. Neural Network with Pruned Filters. US Patent App. 15/590,666
Parallelized Machine Learning With Distributed Lockless Training. US Patent App. 14/875,773